



Peptide neutralises virus

MIT researchers also used simulations to find a protein that binds with and destroys the new virus. <http://digit.in/sep20-57>



Exploring neurological effects

Researchers are trying to understand the affects of the virus, including loss of the sense of smell. <http://digit.in/sep20-58>

can inform healthcare workers if a person has contracted the disease before, even if they were asymptomatic. If these tests are made cheaper and more widely available, they can help track the spread of the disease better, as well as identify plasma donors for treating patients who are suffering from the COVID-19 disease. Serological tests for example were successfully used to contain the spread of the virus in the slums of Dharavi, in Mumbai. Researchers are developing new tests that focus on the spikes of the virus, the protein structures that allow it to enter the human body. At the tip of the spikes, is an area known as the receptor-binding domain which binds with proteins in the human body. This structure is what allows the protein to enter the cells in the airway, and is widely different from the same area in other coronaviruses. By focusing on this domain, researchers hope to develop highly specific antibody tests. Researchers from the University of Michigan have developed a serological test using microfluidic devices, called a lab on a chip, that allows for fast, cheap, portable and precise tests, four factors that rarely come together in a single test. The devices have been tested with synthetic antibodies, and are not being tested with blood from human donors.

If the tests can be made cheap and universally accessible, in such a way that people can test themselves regularly without the need to visit a clinic, then such a test would be as good as a vaccine according to Michal Mina, an epidemiologist from Harvard. The Wyss institute has developed such a cheap paper based test, that costs less than half a USD (about ₹37). The test is not as accurate as nasal swabs and lab based diagnostics, but like a home pregnancy test, can inform people when they are most infectious. Researchers from Purdue University have developed a paper based test as well, where a sample from a nose swab or throat swab are introduced to a small device with the reagents

for detection already mixed in. When the device is folded over, the testing begins, and a result is available after about 40 minutes. Right now the testing device has to be hand made in a lab, but it is possible to use existing production technologies to manufacture these tests at scale as well. This kind of testing method can impede the transmission of the virus. MIT and 3M are collaborating on developing a paper based testing kit as well. The tests are ready, they only need regulatory approval and funding for scaling production.



The test developed by the Wyss institute at Harvard



The paper based testing device from Purdue University

UNDERSTANDING THE VIRUS BETTER

Caltech researchers previously worked with the HIV virus, understanding how the immune system fights HIV, and tackles the virus with antibodies. The same team is now using similar techniques to understand how the immune system responds to the SARS-CoV-2 virus. Using the blood plasma from those who had recovered from COVID-19, the team captured images of antibodies

latching on to an important protein structure on the surface of the virus. The team was also able to identify the weak spots on the virus that the immune system mechanisms were particularly effective against. These were the first ever images of the specific antibodies for neutralising the SARS-CoV-2 virus. The researchers also identified the structures on the receptor-binding domain that the virus uses to infiltrate healthy cells. The antibody would latch on to the virus at this structure, preventing the virus from infecting healthy cells.

At times, the immune response from the body is not great, which is apparent in the patients showing severe symptoms. After getting infected by the virus, some patients are asymptomatic, while others show mild symptoms, and still others show severe symptoms. Researchers from Stanford investigated why this was the case. In the blood of the severely ill patients, the researchers found molecules that promote inflammation. The researchers also found the bits and pieces of pulverised bacteria in the bloodstream, such as bacterial DNA and pieces of cell walls. The more severely ill patients had higher amounts of this bacterial debris in the blood. The study showed that in the patients that are showing more severe symptoms, the bacterial products that are found only in particular parts of the body, such as the lungs or guts, make their way into the bloodstream, which is then pushed throughout the body by the circulatory system. These molecules that promote inflammation, enter the bloodstream from the lungs of the patients, which is the site of the infection. The study found that in these severely affected patients, the immune system is sluggish, and the protective mechanisms of the body act more slowly. The inflammation causing molecules can be the targets of future therapies.

Another team from the University of Utah figured out exactly what goes on in the immune system to cause some people to experience

Credit: Harvard University

Credit: Purdue University photo/Rebecca McElhine